

Poisson distribution

Conditions :

1. Number of trials are large ($n \rightarrow \infty$)
2. Probability of success is very low ($P \rightarrow 0$)
3. Mean ($\mu = \lambda = np$) is finite, positive and moderate.
4. Trials are independent.
5. There are only two events i.e. success and failure (p and q).
6. Events are mutually exclusive

Poisson Distribution - Formula :

1. Mean = $\mu = \lambda = np$
2. Variance = $\sigma^2 = \text{Mean} = np$
3. Standard Deviation = $\sigma = \sqrt{\text{mean}} = \sqrt{np}$
4. $P(x) = \frac{e^{-m} m^x}{x!}, \quad x = 0, 1, 2, 3, \dots, \infty$

Poisson Distribution - Miscellaneous :

1. It has only one Parameter i.e. mean.
It means poisson distribution is uniparametric.
2. It may be uni-model or bi-model.
3. $e = 2.7183$

Poisson approximation to Binomial distribution :

If n , the number of independent trials of a binomial distribution, tends to infinity and p , the probability of a success, tends to zero, so that $\lambda = np$ remains finite, then a binomial distribution with parameters n and p can be approximated by Poisson distribution with parameter $m (= np)$.

In other words when n is rather large and p is rather small so that $\lambda = np$ is moderate then

$$b(x; n, p) \xrightarrow{n \rightarrow \infty} p(x; \mu).$$

The probability distribution of the Poisson random variable X , representing the number of outcomes occurring in a given time interval or specified region denoted by t , is

$$p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots,$$

where λ is the average number of outcomes per unit time

Example 5.17: During a laboratory experiment, the **average number** of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

Solution:

Using the Poisson distribution with $x = 6$, average = $\lambda t = 4$

$$P(X = 6) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X = 6) = \frac{e^{-4} 4^6}{6!} = 0.1041$$

Example 5.19: In a certain industrial facility, accidents occur infrequently. It is known that the probability of an accident on any given day is 0.005 and accidents are independent of each other.

(a) What is the probability that in any given period of 400 days there will be an accident on one day?

(b) What is the probability that there are at most three days with an accident?

Solution: Let X be a binomial random variable with $n = 400$ and $p = 0.005$.

Thus, $np = 2$ and $x = 1$ Using the Poisson approximation,

$$(a) P(X = 1) = \frac{e^{-2} 2^1}{1!} = 0.271$$

$$(b) P(X \leq 3) = \sum_{x=0}^3 \frac{e^{-2} 2^x}{x!}$$

Example 5.20: In a manufacturing process where glass products are made, defects or bubbles occur, occasionally rendering the piece undesirable for marketing. It is known that, on average, 1 in every 1000 of these items produced has one or more bubbles. What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles?

Solution

This is essentially a binomial experiment with $n = 8000$ and $p = 0.001$. Since p is very close to 0 and n is quite large, we shall approximate with the Poisson distribution using

$$\lambda = (8000)(0.001) = 8.$$

Hence, if X represents the number of bubbles, we have

$$P(X < 7) = \sum_{x=0}^6 \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X < 7) = \sum_{x=0}^6 \frac{e^{-8} 8^x}{x!} = 0.31337$$

Example

Contamination is a problem in the manufacture of optical storage disks. The number of particles of contamination that occur on an optical disk has a Poisson distribution, and the average number of particles per centimeter squared of media surface is 0.1. The area of a disk under study is 100 squared centimeters. Find the probability that 12 particles occur in the area of a disk under study

$$\text{Mean} = \lambda = np = 100 \times 0.1 = 10 \text{ particles}$$

$$P(X = 12) = \frac{e^{-10} 10^{12}}{12!} = 0.095$$

(b) The probability that zero particles occur in the area of the disk under study

$$P(X = 0) = e^{-10} = 4.54 \times 10^{-5}$$

(c) **Determine the probability that 12 or fewer particles occur in the area of the disk under study**

$$P(X \leq 12) = P(X = 0) + P(X = 1) + \cdots + P(X = 12) = \sum_{i=0}^{12} \frac{e^{-10} 10^i}{i!}$$

Example

For the case of the thin copper wire, suppose that the number of flaws follows a Poisson distribution with a mean of 2.3 flaws per millimeter.

- (a) Determine the probability of exactly 2 flaws in 1 millimeter of wire.
- (b) Determine the probability of 10 flaws in 5 millimeters of wire
- (c) Determine the probability of at least 1 flaw in 2 millimeters of wire.

(a) $\text{Mean} = \lambda = np = 2.3$

$$P(X = 2) = \frac{e^{-2.3} 2.3^2}{2!} = 0.265$$

(b) $\text{Mean} = \lambda = np = 5\text{mm} \times \frac{2.3\text{flaws}}{\text{mm}} = 11.5 \text{ flaws}$

$$P(X = 10) = e^{-11.5} \frac{11.5^{10}}{10!} = 0.113$$

(c) $\text{Mean} = \lambda = np = 2\text{mm} \times \frac{2.3\text{flaws}}{\text{mm}} = 4.6 \text{ flaws}$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-4.6} = 0.9899$$